



SEQUOIA COOKSTOVES IN BENIN



Document Prepared by Aera Group

On behalf of

SEQUOIA PLANTATION BENIN SA U

Project Title	Sequoia cookstoves in Benin
Version	1.9
Date of Issue	09-July-2024
Prepared By	Aera Group
Contact	info@aera-group.fr 28 cours Albert 1er, 75008 Paris, France

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	4
1.3	Project Eligibility	5
1.4	Project Design	5
1.5	Project Proponent	6
1.6	Other Entities Involved in the Project	6
1.7	Ownership.....	7
1.8	Project Start Date	7
1.9	Project Crediting Period	7
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	7
1.11	Description of the Project Activity	8
1.12	Project Location	10
1.13	Conditions Prior to Project Initiation	12
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks	12
1.15	Participation under Other GHG Programs	12
1.16	Other Forms of Credit.....	12
1.17	Sustainable Development Contributions	13
1.18	Additional Information Relevant to the Project	14
2	SAFEGUARDS	14
2.1	No Net Harm	14
2.2	Local Stakeholder Consultation	14
2.3	Environmental Impact	17
2.4	Public Comments	17
2.5	AFOLU-Specific Safeguards	17
3	APPLICATION OF METHODOLOGY.....	17
3.1	Title and Reference of Methodology	17
3.2	Applicability of Methodology	18
3.3	Project Boundary	19
3.4	Baseline Scenario	21

3.5	Additionality	22
3.6	Methodology Deviations	23
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	23
4.1	Baseline Emissions	23
4.2	Project Emissions	23
4.3	Leakage.....	23
4.4	Net GHG Emission Reductions and Removals	23
5	MONITORING	33
5.1	Data and Parameters Available at Validation	33
5.2	Data and Parameters Monitored.....	33
5.3	Monitoring Plan	37

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project activity aims to provide individuals with improved cookstove (ICS). The stoves will be availed for free in all the districts of Benin: Alibori, Atacora, Atlantique, Borgou, Collines, Couffo, Donga, Littoral, Mono, Ouémé, Plateau, and Zou. The ICS distributed through this project will replace the prevailing inefficient three-stone fireplace or traditional charcoal cookstoves such as charcoal clay cookers and charcoal coal pots that are predominantly used by households across the region. The ICS provides for an efficient transfer of the thermal energy generated through the combustion of biomass towards the cooking pot. There is thus a resultant reduction in both biomass consumption and greenhouse gas emissions.

In addition to the reduction of biomass consumption (and associated reduction in expenses to purchase wood or charcoal – or reduction in time gathering firewood), there will also be an economic benefit in that the ICS will be produced locally in the region and will create jobs which will have a large proportion of female employees.

According to a recent World Bank report on Benin's forest note, at present, one of the main drivers of deforestation is the use of the forested area as fuelwood which provides 46 percent of the energy consumed in the country. While cropland increased from 3.7 to 5.3 million ha, forestland decreased from 7.6 to 5.9 million ha, showing a 22 percent decrease and a deforestation rate of 2.2 percent per year between 2005 and 2015. By comparison, the production of charcoal required 3.37 million tons of fuelwood in 2017 compared to 2.5 million tons in 2010¹.

The project seeks to distribute up to 640,000 ICS to households across Benin. Thanks to carbon finance, the cookstoves will be distributed at a 100% subsidy to users. The project aims to avoid on average 951,875 tCO₂e per year and 6,663,125 tCO₂e over the first crediting period.

Audit Type	Period	Program	VVB Name	Number of years
Validation (crediting period)	20-March-2023 – 19-March-2030	VCS	Applus	7
Total	1	1	1	7

1.2 Sectoral Scope and Project Type

¹ [Benin-Country-Forest-Note.pdf](#)

Sectoral Scope: 03

Project Type: Type II –Energy Efficiency Improvement Project

Grouped Project: No

1.3 Project Eligibility

As per para. 2.1.1 of the VCS Standard, V.4.4 the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Elements	Project applicability
The six Kyoto Protocol greenhouse gases	The project activity aims at CO ₂ reduction of non-renewable biomass.
Ozone-depleting substances.	n/a
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The project applies VMR0006 v1.1 which is a VCS methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	n/a
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	n/a

Furthermore, it cannot reasonably be assumed that the project generates GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. All the households involved in this project activity are fully independent from the project proponent.

The project aims to provide individuals with efficient or improved cookstove (ICS) which falls under the category of efficiency improvements in thermal applications, therefore eligible under the scope of VCS Program. As per VCS Standard Version 4.3, efficiency improvements in thermal applications (e.g., cook stoves) are not excluded, hence project is eligible under VCS.

The stoves will be distributed for free in Benin. Double counting will be prevented by each ICS distributed as part of this project activity having a project identifier and unique serial number inscribed in the metal. This will prevent any other project proponent, organization, retailer, individual or distributors from using the same ICS and including it in another project activity.

1.4 Project Design

The project activity involves the free distribution of ICS devices to households within the project boundary (Benin). All the departments of Benin (Alibori, Atacora, Atlantique, Borgou, Collines,

Couffo, Donga, Littoral, Mono, Ouémé, Plateau, Zou) are included in the project boundary. All the stoves distributed have a unique serial number, and all the cookstoves distributed will be recorded in an electronic database, which will be publicly available to ensure that double counting is avoided. Furthermore, the project will target end-users in the project boundary that are still using inefficient cookstoves where other GHG programs are not operating. Moreover, the projects aim at distributing only one ICS per household;

To account for the continued use of baseline devices, the adjustment factor will be assessed during monitoring period to calculate a $B_{oldadjusted}$ as per methodology requirements.

The distribution of energy efficient cookstove distribution falls under the category of efficiency improvements in thermal applications of non-renewable biomass, therefore it is eligible under the scope of VCS Program.

Eligibility Criteria

The project is a not a grouped project hence this is not applicable.

1.5 Project Proponent

Organization name	Sequoia Plantation Benin SAU
Contact person	Aavaneesh Arja
Title	Project developer
Address	Cotonou, Bénin
Telephone	+22954658210
Email	aavaneesh.arja@arisenet.com

1.6 Other Entities Involved in the Project

Organization name	Aera Group
Role in the project	Carbon advisor and trader
Contact person	Alexandre Dunod
Title	COO
Address	28 cours Albert 1er - 75008 Paris
Telephone	+33 642960978
Email	a.dunod@aera-group.fr

1.7 Ownership

Sequoia Plantation Benin SAU the project owner, has granted full and uncontested legal ownership of the emission reductions that are generated under this VERRA project to Aera Group SAS, and has legal rights concerning changes in use of resources required to service the Project for ownership of emission reductions.

Each ICS have a unique serial number linking it to the project activity and is necessary to establish the fact that the ICS belongs to this VCS project. End users will be informed at the time of ICS installation/distribution that the ICS are distributed for free on account of them generating carbon credits and the fact that these credits belong to project proponent. The end user will be required to sign an agreement that transfers the rights of emissions reduction generated by their cookstoves to Sequoia Plantation Benin SAU.

1.8 Project Start Date

As per the user agreement signed between Sequoia Plantation Benin SA U and Ms. Challa Juliette, project start date is 20-March-2023. It is the date of the first ICS distributed & start of GHG emissions reductions².

1.9 Project Crediting Period

The project crediting period will be seven (7) years – renewable 2 times for a total of 21 years maximum. Starting from 20-March-2023, ending 19-March-2030.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	x

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
20-March-2023 – 31-December-2023	135,500
2024	492,751

² As per VCS standard V4.4; “The project start date of a non-AFOLU project is the date on which the project began generating GHG”

emission reductions or carbon dioxide removals”.

2025	785,465
2026	1,072,613
2027	1,333,279
2028	1,320,405
2029	1,307,401
01-Jan-2030 – 19-March-2030	215,711
Total estimated ERs	6,663,125
Total number of crediting years	7
Average annual ERs	951,875

1.11 Description of the Project Activity

Multiples design of ICS will be distributed for free by the project developer. To date the ICS distributed under the project is OPTIMA-BENIN. They are produced from moulded clay which is dried through high temperatures in a kiln to reinforce its properties including high thermal efficiency and durability, or alternatively dried in a cool, dry, ventilated area out of direct sunlight to cure the final products. It is a simple to use, multi-purpose stove that can be used in a clay, metal cladded or even built into a hearth of a home for more permanent use. Other types of ICS may be also distributed by the project developers. Although exact names and models of other types of stoves is not decided yet. Project developer will ensure that they all comply with the applicability criteria, this will also be checked by VVB at each verification period. Those models are distributed to households, in both rural and urban areas and use exclusively charcoal.

Figure 1: Picture and specifications of the stove

Weight: about 9.5 kg
Dimensions: 24 cm height x 30 cm wide
Cooking power: 3.104 kW
Efficiency: >33%
Durability: 5 years
Fuel: charcoal
Type of household: mobile



Figure 2: ICS design

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / location of ICS household
- Phone number of ICS user or household, where available
- Stove model

- Date of distribution/installation
- ICS serial number
- Type of fuel used.

The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes.

The database will be checked periodically and only one project stove per household will be accounted for emission reduction calculations.

The ICS distributed “OPTIMA-BENIN” have a better efficiency than traditional firewood or charcoal stoves.

OPTIMA-BENIN stove has efficiency of 33.65% thus avoiding GHG emissions linked to the combustion of fuels. The project aims to avoid on average 951,875 tCO₂e per year and 6,663,125 tCO₂e during the first crediting period. The project will target rural and urban, households using whether firewood or charcoal based stoves. Only ICS distributed as replacement of inefficient charcoal and firewood stoves will be accounted for emissions reduction calculation.

1.12 Project Location

The project activity will be located solely within the geographic borders of the republic of Benin. The project activity will occur within the regions of Alibori, Atacora, Atlantique, Borgou, Collines, Couffo, Donga, Littoral, Mono, Ouémé, Plateau, and Zou.

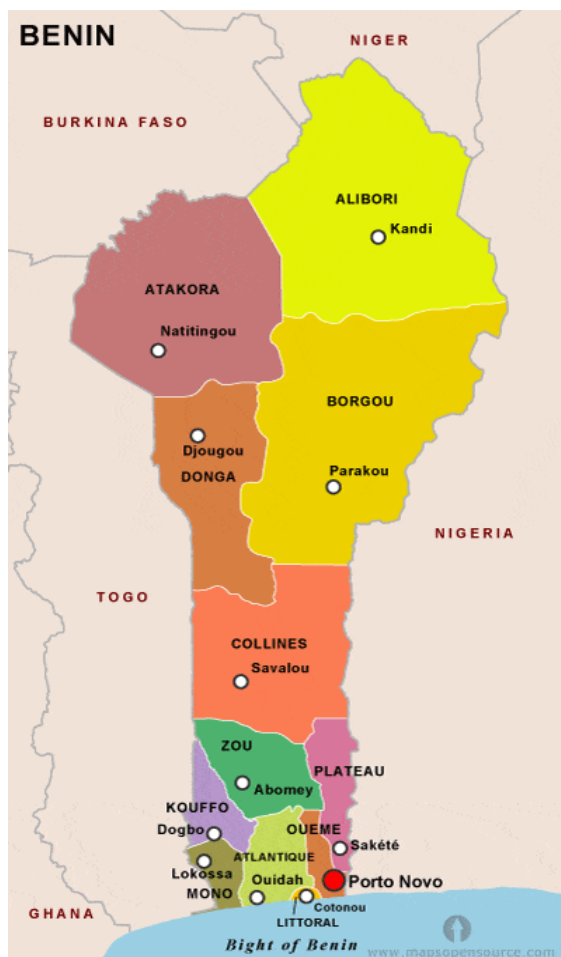


Figure 3: Location of all the regions in which the project will occur.

Region	Geodetic coordinates
Alibori	11° 07' 25.80" N ; 2° 56' 7.79" E
Atacora	10°42'57.79" N 1°31'59.35" E
Atlantique	6°32'26.98" N 2°13'23.14" E
Borgou	9°42'35.08" N 2°44'32.15" E
Collines	8°06'30.77" N 2°11'7.12" E
Couffo	6°48'52.62" N 2°02'25.78" E
Donga	9°16'4.73" N 1°43'1.45" E
Littoral	6°21'58.75" N 2°25'1.42" E
Mono	6°27'27.45" N 1°52'2.67" E

Ouémé	6°36'4.87" N 2°33'2.04" E
Plateau	7°06'0.94" N 2°38'27.29" E
Zou	7°17'0.36" N 2°07'21.94" E

1.13 Conditions Prior to Project Initiation

The majority of households in Benin still rely on woody biomass for cooking. Most households use the charcoal cookstoves or three-stone-fire as their principal method of cooking. The condition prior to the project initiation is the same as the baseline scenario in section 3.4.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There is no regulatory requirements regarding distribution of improved cookstoves within the republic of Benin. A review of relevant laws of Benin did not highlight any specific concerns or requirements around cookstoves or any aspects related to this project activity.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not seeking registration in whole or in part under any other GHG program outside this project activity within VERRA.

1.15.2 Projects Rejected by Other GHG Programs

The project is not seeking registration in whole or in part under any other GHG program outside this project activity within VERRA, and no components have been rejected or excluded from any other GHG emission reduction project.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project activity has no components included in any emissions trading program.

1.16.2 Other Forms of Environmental Credit

The project is not seeking registration in whole or in part under any other form of environmental credit. The project is eligible to participate in carbon registries like - GS4GG, CDM, GCC. However, the current project is not part of the same and will not claim environmental credits from the same.

Supply Chain (Scope 3) Emissions

As per DECEMBER 2022 VCS PROGRAM UPDATE, the public statements required for all emission reductions and removals associated with impacted goods or services will be effective on 1 July 2023. Thus, it is not applicable.

1.17 Sustainable Development Contributions

By reducing the consumption of wood energy for cooking, the improved cookstoves project acts in line with the strategies, programs and projects enabling the preparation and implementation of the NDC in terms of mitigation in the energy sector. In particular, the project to promote low-energy wood technologies described in the report PROJET D'APPUI A LA MISEUVRE DE LA CDN DU BENIN (PROJET N° (PN)/18.2105.7-001.09).

All sustainable development goals will be monitored using project database.

The project activity has a positive impact on sustainable development as outlines below:

ENVIRONMENTAL

Besides Greenhouse gas emissions reduced by the project activity, fuel consumption is reduced for cooking, which reduces the amount of non-renewable biomass use in the region. The project will avoid 6,836,114 tCO₂e during the first crediting period (SDG 13).

Participating households will be more energy sufficient due to the reduction in fuel biomass requirement due to the efficiency of the devices.

Distribute 640,000 high-efficiency, clean burning cookstoves in Benin, to benefit over 640,000 households. (SDG 7.1.2)

Significantly reduce Household Air Pollution (HAP) in rural households (SDG3.9.1).

Thanks to the efficiency of the ICS, wood will be saved and forest areas will be preserved from deforestation (SDG 15)

ECONOMIC

In a region where economic activity and inputs are strained, the annual expenditures on fuel for cooking will be reduced due to the efficient cookstove use, benefiting households to allow spending of resources elsewhere.

Production of the cookstoves regionally will create employment in the region and promote indirect opportunities in the sales, distribution, training and management roles within the supply chain (SDG8).

Additional time will be freed in households from fuel collection – these can now be applied elsewhere in economic or social pursuits.

SOCIAL

Reduce the proportion of time rural women spend on unpaid domestic work related to collection of firewood, and cooking meals on inefficient traditional cookstoves. (SDG5.4.1). Therefore, additional time will be freed in households from fuel collection – these can now be applied elsewhere in economic or social pursuits.

Indoor pollutants – specifically VOCs and particulates are reduced due to more efficient combustion having a positive effect on health. This has been confirmed by the World Health Organization³.

Participating households will be more energy sufficient due to the reduction in fuel biomass requirement thanks to the efficiency of the devices.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information directly related to the project activity has been excluded from the public version of the project description.

Further Information

No further information was deemed to be required for this project activity.

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

Public local stakeholder consultation has taken place on 8, February, 2023. The public consultation was held in the conference room of the hotel "Miracle" located in the town of BOHICON in the department of ZOU.

³<https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

The consultation was organised by the SOCIAL RESEARCH AND ENGINEERING FIRM OF TOGO (CRISTO).

In total, 93 stakeholders from various institutions and beneficiaries attended to the meeting. These actors are composed of members of the Administrative Authorities (Departmental directors of the Living Environment and Sustainable Development, heads of the Forestry Inspection), community leaders (village chiefs and CVD managers), members of members of the Civil Society Organizations (associations/NGOs), and 63 people from the local population⁴.

The stakeholders were informed about the local stakeholder consultation by several ways⁵ :

- Physical submission of invitation letters (for administrative staff, local representatives and representatives and CSO leaders)
- Emails (for administrative staff, local representatives and CSO leaders)
- Telephone calls (for some cookstove users)

Mrs. TOUDONOU Béatrice, Departmental Director of Living Environment and Sustainable Development ZOU welcomed the participants and officially opened the consultation. The Director of CRISTO, Mr. Gabriel BATAWILA, then presented the company SEQUOIA Plantation.

The second presentation was made by Mr. DONGO Jean, CRISTO's statistician-informatician. Several points were addressed during his presentation:

- Human activities that contribute to the emission of greenhouse gases and their short and long term consequences ;
- The current environmental situation in Benin ;
- The objectives, activities, intervention area and targets of the project ;
- The benefits of improved stoves for users, populations and the country).

After the presentation of the project, the organisers provided answers to all participants.

Points raised by different groups during the Local Stakeholders Consultation and answers are provided below:

Folio	Stakeholder	Questions/ contributions	Answers given
1.	VDC representative	We have done a study on wood supply patterns and it is in the cities that wood ends up. In your improved stove supply strategy you will need to focus on where wood is used in mass, especially in the larger restaurants/gargot/cafeterias in cities where the population density is higher.	Indeed, it is in this sense that for the first phase of the project, the areas of intervention are the urban and peri-urban environments.

⁴ The complete list of those who attended is available in Annex 1 "List of attendees to the local stakeholders consultation".

⁵ Sample invitation letter and emails are available in Annex 2 "Means of invitation to the stakeholder consultation"

2.	ABECOB NGO representative	Can SEQUOIA Plantation think of a participatory approach by training people who make traditional (unimproved) stoves to retrain them to make improved stoves? Because in the long run, we should not be satisfied with just distributing improved stoves.	The contract for the manufacture of improved cookstoves is being given to several manufacturers in the area. All entities wishing to engage in the activity can approach SEQUOIA Plantation.
3.	Community leader	How will the beneficiaries be selected? Will the homes be made available to local NGOs and village chiefs who will be responsible for redistributing them to the population?	The improved stoves will be distributed through the village chiefs who will be responsible for mobilizing their populations.
4.	Community leader	Does the company have a plant production site?	Yes
5.	VDC representative	We were told about the benefits of the project. There are no advantages without disadvantages. What will be the impact of the mass production of these improved stoves on the sites used? What will be the impact on the traditional pottery making activity? traditional pottery making?	SEQUOIA Plantation has been officially established according to the laws and environmental policies in force in Benin
6.	VDC member	The stoves will be distributed free of charge, what will be the source of funding to ensure free distribution? As the company is Togolese, are there any partnerships in Benin with the Ministry of Living Environment?	As mentioned in the second presentation, the free distribution of the improved stoves is possible thanks to the funds mobilized by the Carbon Finance. There is also the company SEQUOIA Plantation Benin which is a Beninese company
7.	VDC member	Since the improved cookstoves should first take time to store heat so that its internal device conserves energy, can the improved stoves also be recommended for short duration cooking or quick cooking such as heating water?	Yes, the improved stoves can also be recommended for quick cooking as the heat energy is better channeled to the pot compared to traditional stoves.
8.	Executive Director of the NGO ACTION PLUS	Beyond the mechanism of manufacturing and distribution of improved stoves which is not yet well clarified, the action simply aims at the distribution of improved stoves for certain types of cooking. If we are working in peri-urban areas and these types of stoves distributed cannot be used for various types of cooking or certain activities, for example for salt making, which cannot be done with these improved stoves, has the project thought of broadening the range of types of stoves in order to take into account the variety of types of cooking and activities?	SEQUOIA Plantation Benin is already well established in Benin in accordance with local standards and policies. The distribution of improved cookstoves will be facilitated by village or neighborhood leaders who will be responsible for mobilizing their populations. As SEQUOIA Plantation has opted to always distribute free of charge the improved stoves, only the current format will be manufactured.
9.	VDC member	700,000 stoves will be distributed for free. What happens after that? What happens when the number of cookstoves scheduled to be distributed for free is over? Will you still be distributing free improved cookstoves?	SEQUOIA Plantation has made the choice to always distribute free improved cookstoves. This is possible thanks to carbon finance.
10.	Community leader	What will be the criteria for selecting beneficiaries?	There are no specific criteria for distribution of improved cookstoves. Village or neighborhood leaders will be responsible for mobilizing their populations for the free distribution sessions of improved stoves.

11.	Community leader	How will the collection of the clay that will be used for the manufacture of the improved stoves take place and what are the environmental provisions that have been taken?	SEQUOIA Plantation has been officially established according to the laws and environmental policies in force in Benin. In this sense, the raw material will be collected according to the laws and standards in force in the country and in respect of the environment
-----	------------------	---	--

The outcome of the LSC meeting was more than encouraging. The different stakeholders encouraged and supported the project. No negative impact was detected and all the questions that arose during the meeting were answered back satisfactorily.

In order to document the consultation, a detailed report was made by the team in charge of the consultation named "Stakeholder Consultation Report".

As part of the on-going communication and feedback collection on the use of the stoves, the contacts of the CRISTO team and the SEQUOIA Plantation managers were given to the participants in order to allow them to express themselves if necessary. Furthermore, the participants agreed on the implementation of a digital exchange platform for exchange. The choice was made to create a "WhatsApp group" because of its ease of access. This group will include contacts from the CRISTO team, representatives of representatives of SEQUOIA Plantation and administrative and community representatives in Benin representatives in Benin who will be the main relays to the populations.

2.3 Environmental Impact

No negative environmental impacts have been identified for the project and an environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

Public comments period was open from 10-May to 9-June 2023 and no comment was received.

2.5 AFOLU-Specific Safeguards

Not applicable as this project in a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1

AMS-II.G Energy efficiency measures in thermal applications of non-renewable biomass, Version 11.1

Tools: “TOOL30: Calculation of the fraction of non-renewable biomass” Version 4.0

Tool: “CDM Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities” Version 9.0.

3.2 Applicability of Methodology

Applicability criterion VMR0006 v1.0	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen	The proposed project involves deployment of ICS only in households in Benin, and therefore meets this applicability condition.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	OPTIMA-BENIN stove installed under this project is a single pot wood cookstove that have an efficiency of 33.65% as per the as per the Water Boiling test. The efficiency of the stoves is above 25%. Therefore, this applicability condition is met.
Both ‘Projects’ and ‘Large Projects’ can use this methodology	This project is a large project. The methodology is applicable to both project and large project. Therefore, this applicability condition is met.
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	The source of the biomass is from non-renewable sources in Benin. Based on FAO report, deforestation reached 70,000 ha from 1990 to 2000 and 50,000 ha from 2000⁶. These results are aligned with the WorldBank’s study highlighting the fact that there is a poor governance of the forestry sector with ineffective implementation of forest laws, policies, forest management plans, and protected areas even after the implementation of Law 93-009 of 2nd July 1993 governing forests⁷

⁶ [Évaluation des ressources forestières mondiales \(FRA\) 2020 Bénin - Rapport \(fao.org\)](#)

⁷ [Benin-Country-Forest-Note.pdf](#)

	Therefore, it can be proved that non-renewable biomass has been used in Benin since 31 December 1989.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	Not applicable. The ICS are introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion.
The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	A record keeping system and unique identification of the project device will be used. This will include username, user identification number and address/location of the user's house and distribution date. The record keeping system will ensure that each ICS can be traced to project instances to avoid double counting
The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The project proponent will sign an end user agreement with individual household/user to ensure only PP can claim the emission reduction. A default distribution agreement for end users including the provision that emission reductions generated by the stove are owned by the PP will be provided for each user.

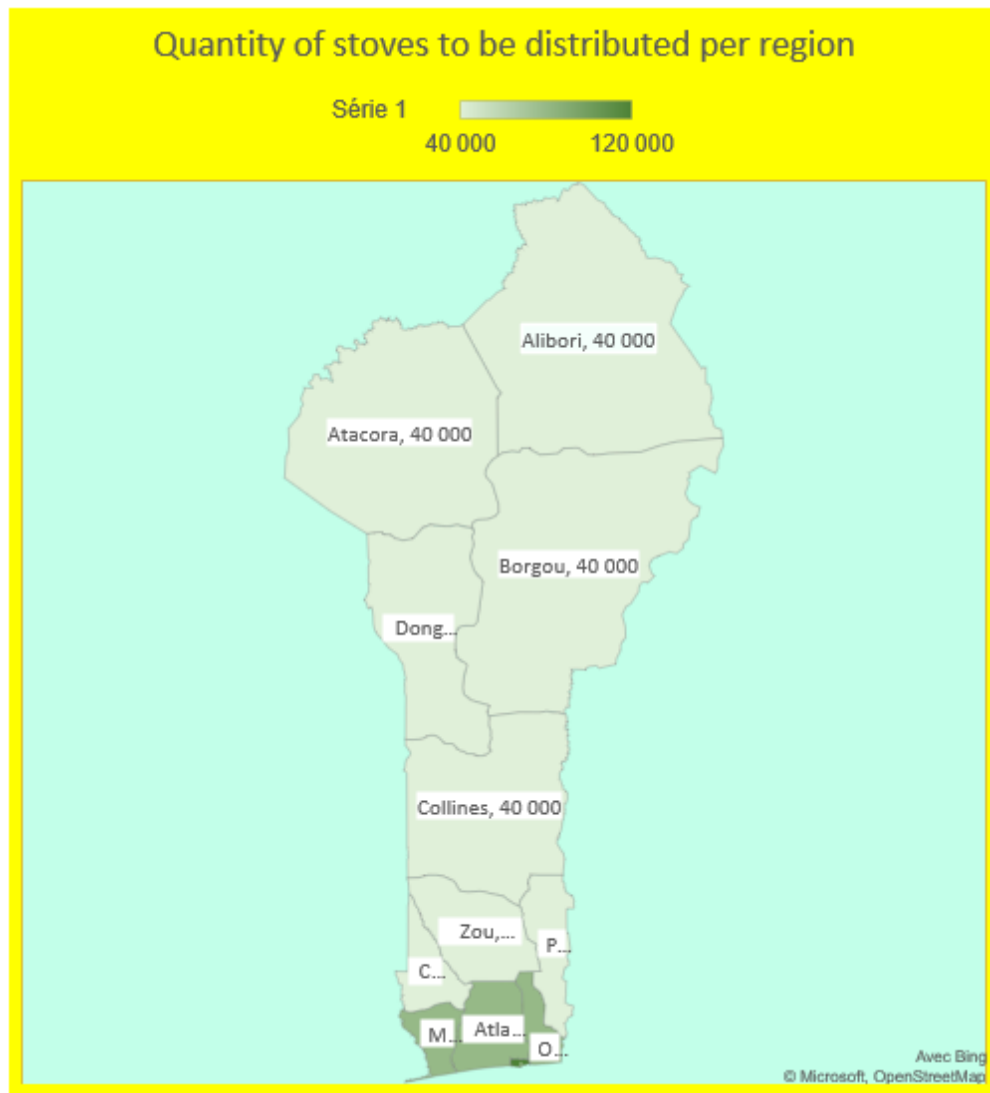
Applicability criterion AMS-II.G v11.1	How the project complies
---	---------------------------------

The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	Each stove has a unique serial number and end users, and each end-user has its own location. All information regarding end-users (names, addresses, telephone numbers) as well as stove identification number is kept in a database and monitored.
The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	At time of distribution, end-users will be informed that the ICS are distributed for free on account of them generating carbon credits and the fact that these credits belong to project proponent. The end user will be required to sign an agreement that transfers the rights of emissions reduction generated by their cookstoves to Sequoia Plantation Benin SAU

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass	CO ₂	Yes	Significant source of emissions
		CH ₄	Yes	Significant source of emissions
		N ₂ O	Yes	Significant source of emissions
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Significant source of emissions
		CH ₄	Yes	Significant source of emissions
		N ₂ O	Yes	Significant source of emissions
		Other	No	No other source identified

The project activity will be located solely within the geographic borders of the republic of Benin. The project activity will occur within several regions, located in the north, west, east and south of the country.



3.4 Baseline Scenario

The baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

For ex-ante calculation, a weighted average value of nold parameter (efficiency of baseline stoves) has been calculated based on the number of charcoal & firewood users.. The value is derived from the baseline survey⁸ conducted by CABINET DE RECHERCHE ET D'INGENIERIE SOCIALES DU TOGO (CRISTO). The survey covered 330 households selected nationwide using the POLITZ method. A total of 615 stoves were found in the 330 households surveyed. Of these 615

⁸ [Baseline report Benin](#)

households, 549 were unimproved cookstoves using charcoal or wood as fuel. Therefore, 89% of the stoves used in households were unimproved wood or charcoal stoves (i.e. 549 out of 615 stoves). Of these 549 unimproved stoves, 294 are unimproved charcoal stoves, while 255 are unimproved wood stoves (3 stones). As default values from methodology for the efficiency of baseline charcoal & firewood stoves are applied (20% for baseline device using charcoal & 10% for baseline device using firewood), the resulting weighted average value of nold is approx. 15%⁹ (50% charcoal users and 50% firewood users).

3.5 Additionality

The methodology uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework. Households may only participate voluntarily in this project.

Step 2: Positive List

This project activity meets all the applicability conditions of the methodology, as demonstrated above, but also the conditions above:

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional => Project proponent will distribute the cookstoves at no cost to the end user and has no other source of revenue other than the sale of GHG credits,
2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology => Project activity is not implemented as part of government schemes nor supported by multilateral funds.

Hence the project activity is deemed additional.

Step 3: Project Method

Not applicable as the project is distributed at zero cost.

⁹ Value to be updated on each verification.

3.6 Methodology Deviations

There is no methodology deviation.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The approved methodology does not account for baseline emissions separately but quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves (see section 4.4. below).

4.2 Project Emissions

The approved methodology does not account for project emissions separately but quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves (see section 4.4. below).

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with Section 5.4 of the approved methodology.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER = \sum_i \sum_j ER_{y,i,j}$$

i	=	Indices for the situation where more than one type/model of improvedcookstove is introduced to replace three-stone fire
j	=	Indices for the situation where there is more than one batch of improvedcookstove of type i

ER_y	= Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	= Emission reductions by improved cookstove of type <i>i</i> and batch <i>j</i> during year y in t CO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where:

$B_{y,savings,i,j}$	= Quantity of woody biomass that is saved in tonnes per improved cook stove of type <i>i</i> and batch <i>j</i> during year <i>y</i>
$f_{NRB,y}$	= Fraction of woody biomass that can be established as non-renewable biomass (fNRB)
$NCV_{wood\ fuel}$	= Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)
EF_{wf,CO_2}	= CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel,
$EF_{wf,non\ CO_2}$	= Non-CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ)
$N_{y,i,j}$	= Number of improved cook stoves of type <i>i</i> and batch <i>j</i> operating during year <i>y</i>

$$B_{y,savings,i,j} = B_{old} \times (1 - (\eta_{old} / \eta_{new,i,j}))$$

$$\text{With } \eta_{new,i,j} = \eta_P \times (DFn)^{y-1} \times 0.94$$

Where:

B_{old}	= Quantity of woody biomass that is saved in tonnes per improved cook stove of type <i>i</i> and batch <i>j</i> during year <i>y</i>
η_{old}	= Efficiency of baseline cookstove
$\eta_{new,i,j}$	= Efficiency of the improved cookstove type <i>i</i> and batch <i>j</i> determined through water boiling test (WBT) during year <i>y</i> Alternatively, efficiency may be determined using Equation 4
η_P	= Efficiency of project stove (fraction) at the start of project activity,
$(DFn)^{y-1}$	= Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards Alternatively default value of 0.99 efficiency loss per year can be considered

0.94	= Adjustment factor to account for uncertainty related to project cookstove efficiency test
------	---

Parameter	Value	Unit	Description of the parameter	Source/reference/justification
EF_{wf,CO_2}	112	tCO ₂ /TJ	CO ₂ emission factor for the use of wood fuel in baseline scenario	IPCC default for wood fuel,
$EF_{wf,non\ CO_2}$	26.23	tCO ₂ /TJ	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario	IPCC default for wood fuel
$N_{y,i,j}$	640,000	Cookstoves	Number of cookstoves distributed per year over the project lifetime	Ex-ante estimation of operating cookstoves
$NCV_{wood\ fuel}$	0.0156	TJ/tonne	Net calorific value of the non-renewable woody biomass that is substituted or reduced	IPCC default factor for wood
η_{old}	15% (Ex-ante)	fraction	Efficiency of baseline stoves	Weighted average value based on type of fuel of cookstoves which will be replaced (50% firewood based stoves, 50% charcoal based stoves)from VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, v1.1
$\eta_{new,i,j}$	33.65% (1 st year)	fraction	Efficiency of the improved cookstove type i and batch j. Efficiency may be determined using Equation 4	Result from WBT

f_{NRB}	76.22%	Fraction	Fraction of woody biomass that can be established as non-renewable biomass	Calculated
B_{old}	3.624	tonnes	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove	Calculated (section 5.1)
η_p	33.65% (1 st year)	Fraction	Efficiency of project stove (fraction) at the start of project activity	Obtained from WBT
$(DF_n)^{y-1}$	0.99	-	Discount factor to account for efficiency loss of project cookstove per year of operation(fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered	Default value from VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, v1.1
$B_{y,savings,l,j}$	1.61 (1 st year)	tonnes	Quantity of woody biomass that is saved in tonnes per improved cook stove of type	Calculated
μ_y	80%	Fraction	Adjustment to account for any continued use of pre-project devices during the year y	Ex-ante estimation

L	95%	fraction	Leakage adjustment factor	Methodology default
ER	6,663,125	tCO ₂ e	Annual average emissions reduction	Calculated as per methodology equation

The quantity of firewood consumed in absence of project activity (**B_{old}**) shall be determined using an estimation of average annual consumption of firewood per household which may be derived using any of the following options:

- (a) Historical Data. Project proponent shall ensure that the relevance of data is appropriately justified for the target population and is the latest available data from credible source(s).
- (b) Baseline Survey of Local Usage. Project proponent shall carry out a survey of usage prior to implementation of the project activity following the sampling approach described in the latest version of CDM document Sampling and surveys for CDM project activities and programme of activities. Alternatively, the project participant may follow the simple random sampling approach and the minimum sample size should be determined as per the following guidelines:
 - Project target population < 300: Minimum sample size 30
 - Project target population 300 – 1000: Minimum sample size 10% of group size
 - Project target population > 1000: Minimum sample size 100

This simplified approach may also be used for determining minimum sample size for parameters listed under Sections 9.1 and 9.2 in which case it is not requisite for the sample size to meet confidence/precision requirements

- (c) Minimum Service Level. Where historical data or a baseline survey has not been conducted, a default value of 0.5 ton/capita/year may be considered as the baseline biomass consumption. Household size shall be determined using credible references/literature or target population specific surveys. The survey shall be conducted as per guidelines outlined in option (b) above

The equations below shall be used for calculating biomass consumed in absence of project activity in case more than one project stove is used in household:

$$B_{old,i,j} = B_{old,HH} + N_{d,HH}$$

$$B_{old,HH} = B_{old,p} \times N_{p,HH}$$

Bold, p	Annual quantity of woody biomass that would have been used per person in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/person/year).
Np, HH	Average number of households
Nd,HH	Number of project devices per household

Bold, HH

=Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/household/year)

As per section 8.4 of the applied methodology, Bold can be estimated using Historical Data of (Option a). This option is chosen for the determination of Bold. Therefore, above parameters are determined using historical data.

The consumption of woody biomass is retrieved from most recent and credible data available from Directorate General of Energy Resources..

Determination of Bold,p

The consumption of woody biomass is retrieved from the result of the study conducted by Ministry of Energy of Benin called “*KEY FIGURES 2021 ENERGY BALANCES AND INDICATORS 2016 TO 2020*”¹⁰, which describes Benin’s fuel consumption habits.

The result of the study shows a consumption of 0.697 tonnes/person/year for wood fuel .

Therefore, data from this study as most recent will be used for Bold,p.

Determination of Np,HH:

Based on the latest data from National Institute of Statistics and Economic Analysis (INSAE)¹¹ showing households’ size and composition and which was published in 2019, the average household size is 5.2 for Benin.The NGO “CABENT DE RECHERCHE ET D’INGENIERIE SOCIALES DU TOGO (CRISTO)” also conducted a baseline survey specific for this project and found that the average household size is 6.3. Thus, in order to be conservative, the value of 5.2 from INSAE was used.

Thus, this value was selected for household’s size estimation.

Determination of Nd,HH:

As the project is expected to distribute only one ICS per household, therefore the Nd,HH value is 1.Determination of Bold,HH:

$Bold, HH = Bold, p \times Np, HH = 0.697 \times 5.2 = 3.624$ tonnes per household and per year.

Determination of Bold,HH adjusted:

$Bold,HH \text{ adjusted} = Bold, i, j \times (1 - \mu_y) = 3.624 \times 80 = 2.900\%$

¹⁰ <https://aera.cloudvault.m-files.com/SharedLinks.aspx?accesskey=d8855b96e7f58c6ed217eba4d621263bad85f186a74a66961b1e0f51f82740f2&VaultGUID=ffebd84a-a981-4720-85e2-a323c816e47e>

¹¹ <https://dhsprogram.com/pubs/pdf/FR350/FR350.pdf>

Determination of $B_{old,i,j}$:

$B_{old,i,j} = B_{old,HH} \div N_{d,HH} = 3.624 / 1 = 3.624$ tonnes. All the parameters are resumed in the table below:

Parameters	Value	Source
$B_{old,p}$	0.697 tonnes/person/year	Calculated
$N_{d,HH}$	1	Ex-ante estimation
$N_{p,HH}$	5.2	Study result
μ_y	80%	Ex-ante estimation
$B_{old,HH}$	3.624	Calculated as per methodology VMR0006 v1.1
$B_{old,HH} \text{ adjusted}$	2.900	Calculated as per methodology VMR0006 v1.1

Determination of efficiency of ICS during year y :

As per equation 5 of methodology applied $\eta_{new,i,j} = \eta_P \times (DF_n)^{y-1} \times 0.94$

The ex-ante estimated efficiencies are therefore:

Year	$\eta_{new,i,j}$
20-03-2023–31-12-2023	33.65%
2024	31.6%
2025	31.3%
2026	31.0%
2027	30.7%
2028	30.4%
2029	30.1%
01-01-2030–19-03-2023	29,8%

Determination of $B_{y,savings}$. $B_{y,savings,i,j} = B_{old} \times (1 - (\eta_{old} / \eta_{new,i,j}))$

Year	Bold adjusted(tonnes)	$\eta_{new,i,j}$	η_{old}	$B_{y,savings,i,j}$ (tonnes)
------	-----------------------	------------------	--------------	------------------------------

2023	2.900	33.65%	15%	1.61
2024	2.900	31.6%	15%	1.52
2025	2.900	31.3%	15%	1.51
2026	2.900	31.0%	15%	1.50
2027	2.900	30.7%	15%	1.48
2028	2.900	30.4%	15%	1.47
2029	2.900	30.1%	15%	1.45
01-01–20230– 19-03-2030	2.900	29,8%	15%	1.44

Determination of fNRB :

fNRB was estimated following tool 30 *Calculation of the fraction of non-renewable biomass* V4.0 methodology : The values obtained is 76.22%. Main results for calculation are provided below.

Parameter	Value	Source of data
NRB	3,050,589 t/yr	Calculated
RB	951,699 t/yr	Calculated
H	4,002,288 t/yr	Calculated
MAIforest,i	1.24 t/ha/yr	Calculated
MAIother	1.24 t/ha/yr	Calculated
Fforest	3,135,150 ha	FRA2020 country report ¹²
fNRB	76.22 %	Calculated

Applicability of para 13 of TOOL 30 v4.0:

As per paragraph 10 of the tool, the parameter fNRB shall be estimated using the most recent historical year for which data is available. For the fNRB assessment, the values for Fforest,i, Fother,i, Pforest, and biomass consumption have been assessed using the most recent data from the FAO, specifically the Forest Resources Assessment 2020 and the FAO Yearbook of

¹² <https://fra-data.fao.org/assessments/fra/2020/BEN/sections/extentOfForest> (table: Extent of forest and other wooded land, Forest (a), 2020.)

Forest Products 2020. However, for above-ground biomass growth and the distribution of forest areas by ecological zones, the data have been sourced from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and the FAO Global Forest Resources Assessment 2000, respectively. Although these data are not from 2020, they are the latest available data for these parameters. Therefore, these values have been used for the fNRB assessment.

As per paragraph 13, the value of estimated NRB has been compared with the product of total average above ground biomass and most recent available observed annual rate of deforestation. The result shows that the estimated NRB value is less than 10% above the value calculated as per the product of biomass and deforestation rate. Thus, a justification not required¹³.

Comparison of calculated fnrb with values from literature:¹⁴

The achieved fNRB was compared with those available on literature mentioned by tool 30 version 4 (Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change. 5(3), pp. 266–272) however, the fnrb from the literature was not used for the following reasons:

The study was made in 2015, which means 8-9 years ago. And just as for other parameters, a value that old is not taken into account, unless it cannot be found via another source or calculated. data used to calculate fNRB results from country-driven official reports and statistics, just as the global forest resources assessment, which data is provided by a well-established network of officially nominated national correspondents.

Data used for wood consumption and local supply/demand is from 2009, which makes these value non applicable to the project as the values at this time may not represent actual situation. Furthermore, since 2015, the Beninese population continued to increase meaning a constant increase of wood fuel consumption. As per latest available data from Forest resource assessment¹⁵ forest area continues to decrease (-1 million ha between 2000 and 2020). In the same time, firewood is still the main fuel source used by households for energy¹⁶.

The study conducted by Bailis and al, is based on the WISDOM approach that rely on extensive data research, geo-spatial analysis, hypothesis based on researchers' knowledge, while the tool 30 is a more accessible way of measuring the fNRB, making it difficult to compare both results.

Also, the study explains that "For analytical purposes, the continuous map resulting from the weighted interpolation analysis is segmented into buffers; cities with high demand produce wide woodshed buffers and cities with low demand produce narrow buffers, which simulates the territory under pressure from urban and high-deficit rural areas. Woodsheds within each national boundary are defined by using zonal

¹³ Estimated NRB = 3,050,589 and the product of Above ground biomass * Annual deforestation rate = 5,178,000. Thus, the difference is less than 10% (See ER sheet for detailed calculation).

¹⁴ The calculated fnrb was compared with fnrb from the study conducted by Bailis et al. 2015

¹⁵ <https://fra-data.fao.org/assessments/fra/2020/BEN/sections/extentOfForest/>

¹⁶ <https://au-afrec.org/benin>

statistics to calculate the supply/demand balance of each buffer” However, buffering points can underestimate or overestimate woodshed availability in each zone, based on statistics used. As explained in the study, this approach assumes optimal harvesting of DEB (dendro-energy biomass). It does not reveal actual harvesting patterns” Thus it could lead to inaccurate values for each zone and buffer.

On the other hand, data used to calculate fNRB results from country-driven official reports and statistics, just as the global forest resources assessment, which data is provided by a well-established network of officially nominated national correspondents.

As no new study conducted following WISDOM to assess fNRB have been found in literature, and as reassessing the fNRB with WISDOM, using latest data available requires both time, but also some high level technical knowledge that are well above what is requested by tool 30 to assess the fNRB, project developer can neither use outdated values from the report nor reassess the value based on WISDOM for Benin.

Conclusion:

The fNRB assessment methodology used by Bailis and al; does not seems to be applicable to this project, hence the results cannot be used for the project. The report does not use the CDM Tool 30 to assess the fNRB although this is the tool proposed by both AMS II-G and VMR0006 (fNRB default values from CDM) to assess the value of fNRB. Moreover due to the fact that these values from the study rely on really old data, with no guidance from VERRA on how to reassess fNRB based on WISDOM methodology with updated data, project developer tend to believe that although many fNRB assessments methods exist (tool 30, WISDOM, MoFuSS) with both their pro and cons, the most used and accepted so far way of assessing the fNRB (cdm tool 30) can be used and value assessed by project developer can be applied

The following table shows the details of the net GHG emission reductions per year:

<i>Estimated greenhouse gas emission reductions per year:</i>	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
20-March-2023 to 31-Dec-2023	135,500	0	0	135,500
01-January-2024–31-12-2024	492,751	0	0	492,751
01-January-2025–31-December-2025	785,465	0	0	785,465

01-January-2026–31-December-2026	1,072,613	0	0	1,072,613
01-January-2027–31-December-2027	1,333,279	0	0	1,333,279
01-January-2028–31-December-2028	1,320,405	0	0	1,320,405
01-January-2029–31-December-2029	1,307,401	0	0	1,307,401
01-Jan-2030 to 19-March-2030	215,711	0	0	215,711
Total	6,663,125	0	0	6,663,125

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	Bold,p		
Data unit	tonnes/year		
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices.		
Source of data	Historical data		
	Biomass consumption by capita (t/yr)	0.697	Chiffres-Cles-SIE-2021-vf2 (4).pdf (Total consumption of bioenergy per capita (ton/capita/year))

			page 17) ¹⁷
Value applied	0.697		
Justification of choice of data or description of measurement methods and procedures applied	Based on approximate value of woodfuel consumed by person/year times the average numbers of people per household		
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device		
Comments	These data are further supported by conservative study conducted by NGO CRISTO ¹⁸ which resulted in:		
	Number of person by households	6.3	CRISTO's Survey page 7
	Biomass consumption by capita (t/yr)	0.834	Calculated from study result

Data / Parameter	Np, HH
Data unit	Number
Description	Average number person per households (household size)
Source of data	Data from National Institute of Statistics and Economic Analysis (INSAE) (page 9) ¹⁹
Value applied	5.2
Justification of choice of data or description of	This value was chosen as it more conservative, knowing that the baseline survey conducted specifically by the NGO CRISTO for

¹⁷ <https://aera.cloudvault.m-files.com/SharedLinks.aspx?accesskey=d8855b96e7f58c6ed217eba4d621263bad85f186a74a66961b1e0f51f82740f2&VaultGUID=ffebd84a-a981-4720-85e2-a323c816e47e>

¹⁸ <https://aera.cloudvault.m-files.com/SharedLinks.aspx?accesskey=70a087864ba387fd942213f9e50d0d8d30ccce73f934d86d12f9245f5fde2ffa&VaultGUID=ffebd84a-a981-4720-85e2-a323c816e47e>

¹⁹ <https://dhsprogram.com/pubs/pdf/FR350/FR350.pdf>

measurement methods and procedures applied	this project is less conservative (household size is 6.3). Thus, to be conservative, INSAE's value was applied 5.2. ²⁰ (page 9)
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	n_p
Data unit	Fraction
Description	Efficiency of the project stove at the start of project activity
Source of data	Manufacturer's specification based on Water Boiling Test conducted.
Value applied	0.33
Justification of choice of data or description of measurement methods and procedures applied	Parameter determined ex-ante. Based on water boiling test conducted by the company "SOS ENERGIE TOGO (SOSET)", which carried out tests on the technical performance of three (03) stoves with ceramic hearths, using charcoal fuel, made by a beninese company called "OPTIMA". The tests were carried out according to the WBT 4.2.3 protocol published by the Clean Cooking Alliance on March 19, 2014 and according to the "Guidelines for Testing Charcoal Stoves with WBT 4.2.2" dated June 14, 2013.
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	$f_{NRB,y}$
Data unit	Percentage
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Fraction of non-renewable biomass is calculated as per the f_{NRB} calculation TOOL30, Calculation of the fraction of non-renewable biomass v4: =

²⁰ <https://dhsprogram.com/pubs/pdf/FR350/FR350.pdf>

Value applied	76.22%
Justification of choice of data or description of measurement methods and procedures applied	The fraction of non-renewable biomass was calculated using official data from Global forest assessment from FAO and IPCC values.
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	NCV _{biomass}
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	Default as per applied approved methodology VMR0006
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	EF _{wf,non CO2}
Data unit	tCO2/TJ
Description	CO2 emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006
Purpose of Data	Emission reduction calculations

Comments	This parameter shall remain fixed for the monitoring periods.
Data / Parameter	EF _{wf,CO2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Default methodology value.
Purpose of Data	Emission reduction calculations
Comments	This parameter shall remain fixed for the monitoring periods.

5.2 Data and Parameters Monitored

Data / Parameter	N _{y,i,j}	
Data unit	Number	
Description	Number of project devices of type i and age a that are operating in year y	
Source of data	ICS users' survey	
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.	
Frequency of monitoring/recording	At least once every two years	
Value applied	Year	N _{yi}
	2023	80 000
	2024	230 000
	2025	370 000
	2026	510 000

	2027	640 000
	2028	640 000
	2029	640 000
	2030	640 000
Monitoring equipment	Monitoring surveys	
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter	
Purpose of data	Calculation of baseline emissions	
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value	
Comments		

Data / Parameter	$\eta_{old,i,j}$
Data unit	Percentage
Description	Efficiency of the system being replaced (Traditional Cooking Stoves)
Source of data	Default value from methodology VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, v1.1 (conservative)
Value applied	15%
Justification of choice of data or description of measurement methods and procedures applied	The value of 0.15 is used conservatively for ex-ante estimation as the project will replace stoves using both firewood and charcoal. Thus for ex-post emission reduction calculation, weighted average values regarding the type of stoves replaced will be used.
Purpose of Data	Calculation of baseline emissions
Comments	As option (e) is used for pre-project stoves efficiency, the average value of 15% is used as weighted average value, knowing that the project will replace fireplaces using charcoal (50% of households) and those using wood (50% of households). This parameter will be updated during each monitoring periods Based on (e) the weighted average values as more than 1 type of devices are used as baseline device.

Data / Parameter	$\eta_{new,y,i,j}$
------------------	--------------------

Data unit	Fraction																		
Description	Efficiency of the improved cookstove type i and batch j																		
Source of data	Water Boiling Test (SOSET report)																		
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, v1.1: "Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"																		
Frequency of monitoring/recording	Annual																		
Value applied	<table border="1"> <thead> <tr> <th>Year</th><th>Efficiency</th></tr> </thead> <tbody> <tr><td>1</td><td>33.65%</td></tr> <tr><td>2</td><td>31.6%</td></tr> <tr><td>3</td><td>31.3%</td></tr> <tr><td>4</td><td>31.0%</td></tr> <tr><td>5</td><td>30.7%</td></tr> <tr><td>6</td><td>30.4%</td></tr> <tr><td>7</td><td>30.1%</td></tr> <tr><td>8</td><td>29.8%</td></tr> </tbody> </table>	Year	Efficiency	1	33.65%	2	31.6%	3	31.3%	4	31.0%	5	30.7%	6	30.4%	7	30.1%	8	29.8%
Year	Efficiency																		
1	33.65%																		
2	31.6%																		
3	31.3%																		
4	31.0%																		
5	30.7%																		
6	30.4%																		
7	30.1%																		
8	29.8%																		
Monitoring equipment	n/a																		
QA/QC procedures to be applied	n/a																		
Purpose of data	Calculation of baseline emissions																		
Calculation method	As per equation 5 of VMR0006 i.e. $\eta_{new,i,j} = \eta_P \times (DFn)^{y-1} \times 0.94$																		
Comments	--																		

Data / Parameter	μy
Data unit	Fraction

Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	This parameter should be monitored using one of the following methods: a) If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and Bold, adjusted is equal to Bold. b) If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves. For example, if there were 3 baseline cookstoves in a household and it was determined during the survey that use of one of them continues during the crediting period then a conservative adjustment factor of 0.33 is applied to Bold. Another example would be the case where there was only one baseline cookstove per household and its use during the project period continues along with the improved cookstove to meet 25% of the cooking needs of the household in which case the adjustment factor will be 0.25. Another example would be to interview the household and have them estimate the time of usage of the baseline cookstoves and improved cookstove on an average day.
Frequency of monitoring/recording	Once every 2 years
Value monitored	20% (ex-ante estimation)
Monitoring equipment	Monitoring surveys
QA/QC procedures to be applied	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual /biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. In the case the desired precision is not met, lower bound values shall be used against repeating the survey to determine the operational fraction of stoves.
Purpose of the data	Calculation of baseline emissions

Calculation method	n/a
Comments	The proportion of continued use of pre-project stoves is a monitored parameter, i.e. only available ex-post. For ex-ante purposes, a conservative value of $y = 20\%$ is considered²¹. For the subsequent monitoring sessions of the same project, the previously monitored values shall be used for ex-ante purposes.

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Lifetime based on feedback from users collected by SOS ENERGIE TOGO (SOSET), the company which oversaw the efficiency test of Sequoia improved cookstoves.
Frequency of monitoring/recording	Once at the time of project stove installation
Value monitored	5
Monitoring equipment	n/a
QA/QC procedures to be applied	n/a
Purpose of the data	Calculation of baseline emissions
Calculation method	n/a
Comments	--

Data / Parameter	Date of commissioning of batch j
------------------	----------------------------------

²¹ Consultant has carried out similar projects in other countries and based on the field experience of these projects this value is found consistent.

Data unit	Date	
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.	
Source of data	Project database	
Description of measurement methods and procedures to be applied	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch	
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch	
Value monitored	Batch	Date of commissioning
	2023	20/03/2023
Monitoring equipment	n/a	
QA/QC procedures to be applied	n/a	
Purpose of the data	Calculation of baseline emissions	
Calculation method	n/a	
Comments	--	

5.3 Monitoring Plan

The monitoring plan is designed to monitor the parameters listed in Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey. The share of operating stoves and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. The PP will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

No monitoring for leakage is required, as the Emission reduction will be multiplied by a factor of 0.95 to account for leakage.

Based on CDM Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0, the sampling plan is the following: (a)Sampling design: Due to the large

number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each batch of ICS. Samples will be drawn from ICS recorded in the project database administered by the project proponent.

Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities: “When biennial inspection is chosen a 95 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampling parameter. On the other hand, when the project proponent chooses to inspect annually, a 90 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampled parameter”. Therefore, for parameters monitored annually, sample size should permit to achieve a 90/10 precision, and for parameters monitored biennially, the sample size should be determined to achieve a 95/10 precision.

Monitored Parameters: The project involves introduction of intermediate improved cook stoves as energy efficiency measure:

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating in year y	Proportion of ICS still in operation	Proportion	Biennially
2	$\eta_{new,y,i,j}$ Efficiency of the device of each type i and batch j implemented as part of the project activity	Thermal Efficiency of operational ICS	Mean value	Annually (The project opts to determine the efficiency using the equation given by methodology, therefore it is not required to monitor this parameter via sampling survey.)

3	<i>μy</i> Adjustment to account for any continued use of pre-project devices during the year y	Adjustment factor	Proportion	Biennially
---	---	-------------------	------------	------------

To be conservative, cookstoves that will end their expected lifespan of 5 years, will be removed from the database even if they are still found working. However, cookstoves that will be replaced during their life span will have their life span extended to 5 years starting from the date of replacement ie a cookstove distributed on 1st January 2023 and that should be removed on 31st December 2027, will have its life span extended to 30 December 2032 if the stove is replaced on 31st December 2027. Both date of first installation and date of replacement will be available in the project database for cross-check by the VVB.Target Population

The target population will be the complete number of devices (ICS) distributed under the project activity.

Sampling Method

The project involves distribution of ICS throughout the project area thereby replacing traditional cooking devices in the selected households. The population is heterogeneous in nature i.e., common technology with similar operating characteristics but spread across the eastern parts of the country.

The populations of each batch will be combined, the sample size is determined, and a single survey will be undertaken to collect data. To ensure the survey result is representative of the entire population, the dissimilarity (such as ICS type, vintage and project area in which they are operating) within the project will be taken into account in the sample size calculation. Samples will be drawn by using the random number generator.

Monitoring organization:

The project operator Sequoia Plantation Benin SAU takes the responsibility of the monitoring plan implementation, supervision of the monitoring process, periodic users surveys, data collection and recording, and reporting.

The information collected will be stored in the electronic database available with PP and it will be updated on ongoing basis by trained staff. The database will be sortable by the information collected as per Registration Process and will be made available to the VVB at verification. The PP will verify accuracy and completeness and confirm that there is no double entry of serial

numbers in the database. The PP will identify any discrepancy and the correct information will be entered into the database.

All partners will be required to conform with project design and monitoring systems designed which will cover the above-mentioned role and responsibilities.

All technical staff responsible for installation and maintenance of the stoves will be trained in terms of the understanding the requirements of VCS on the monitoring system. The technical, operational and maintenance trainings provided for the personnel will be described in each monitoring report.

Below, the team in charge of the monitoring plan:

Role	Responsibilities	Identity
Monitoring manager	- In general, ensure that the project follow the monitoring plan - Ensure that the equipment and measurements are in line with the measurement methods, recording frequency and archiving approaches in monitoring plan - Ensure that monitoring data collected is consolidated and entered in electronic database - Ensure that monitoring team receives proper training - Ensure a coherent and standard monitoring report for the verification	SIRANGI Rajesh
Monitoring surveys	- Conduct the monitoring surveys, and follows up visits	Daniel Loconon HOVOZOUNKOU Mahugnon Grace
Data manager	- Collect monitoring data. - Enter data in electronic database and archive hardcopies.	ADDA Roland Fréjus
Consultant	- Carry out sample size determination and emission reduction calculations	AERA Group